

Analysis of the Geometrical Features Detectability Constraints for Laser-Scanner Sensor Planning

E. Zussman, H. Schuler and G. Seliger

IWF – Institute of Machine Tools and Manufacturing Technology, Technical University of Berlin, Berlin, Germany

This paper presents a planning approach for automatic synthesis of the viewpoint locations of a laser scanner that satisfy the detectability constraints of the geometrical features of a given object. Dominant constraints taken into consideration were the laser scanner field of view, depth of view, and the maximum permissible view angle. The approach is task driven and sensor model based. Information is given about the scanned object and its selected features, as well as about the optical characteristics of the sensor. Planning is presented as an optimisation problem where the optimal laser-scanner viewpoint must satisfy all the constraints simultaneously. The approach is demonstrated by scanning a 2D feature, and through planning an experiment in which robotic assembly of a car door is assisted by a laser scanner.

Keywords: Assembly; Laser-scanner; Sensor planning

1. Introduction

A critical aspect of current production technology is the automatic assembly process. Yet even when flexible automation systems are installed it is still difficult to achieve the full potential of customised mass production. The number of failures and the operating costs are high, and the cycle-time for designing and planning the assembly tasks is lengthy. More reliable and efficient assembly automation can be achieved through the integration of sensors into assembly systems. Such systems are equipped with various types of sensor in order to interact in an intelligent and flexible manner with the environment. However, reaching the desired levels of flexibility and intelligence entails planning the way in which the sensors are integrated into the system as well as selecting the appropriate type, number and location of the sensors. Sensor planning, which is the subject of this paper, seeks to

automate requirements such as these in order to facilitate the use of flexible, sensor-based assembly systems without substantial human involvement.

This work is concentrated on sensor planning techniques for laser scanners which can be classified as vision sensors (a detailed sensor model is given in Section 2.2). The closest area of research deals with object feature detection. Where the aim is to find out the sensor parameters, the features of a known object in a known location should satisfy particular constraints when imaged: field of view, depth of view, and view angle. Previous intelligent sensor planning approaches that focused on feature detection fall into three categories: generate-and-test, simulation, and synthesis. In the generate-and-test approach, an appropriate location is often determined by a search in the sensor configuration space guided by specific heuristics. Locations for the sensors are chosen and then tested in order to verify that they meet the requirements of the specified task [1, 2]. The next approach simulates the scene based on the object description and the sensor parameters. The simulation provides a tool for easily generating and testing different sensor locations [3–5]. The synthesis approach is based on analytic determination of the task constraints resulting in the required sensor parameters [6], or even the optimal parameters [7].

In this paper, a synthesis approach is presented in which the optimal viewpoint location of a laser scanner sensor is planned. The particular optical constraints of this sensor and the geometrical constraints of the assembly task differentiate this approach from the above approaches. Section 2 presents the overall approach for integrating sensor planning into assembly process planning. Section 3 presents the sensor model and the object model. The different detectability constraints are developed in Section 4. Section 5 presents the formulation of the merging of the constraints as an optimisation problem. The implementation of the algorithms as well as an example and experiment are given in Section 6.

2. The Approach

The goal of this research is to support and enhance the use of sensors, specifically of laser scanners, for assembly tasks

that are performed by robots. A further goal is to create planning guidelines for determining the optimal configuration of the sensors and the way in which they should be integrated into the assembly task. The approach is task-driven and model-based. Information about the objects exists in a CAD model, while the parameters of the laser scanner are characterised in a specific model. The sensing requirements are determined by making use of image simulation and sensor simulation. The overall goal is to find the optimal configuration of the sensors, that is, the minimum number of sensing points and the minimum error.

The different modules of sensor process planning are described in Fig. 1. The input to the process is the task definition, which specifies the exact features of the object that is to be scanned. Specification of these features is based on the parameters of critical assembly features, such as tight tolerances. Once the features are marked, the feature detection planning module, which is the core of this paper, plans the viewpoint location of the sensors. The planning process integrates the laser scanner parameters and the object features and results in the optimal location of the sensors.

Image and sensor simulation is a statistical characterisation of object variability as a basis for the realistic simulation of images incorporating part-to-part position, surface appearance variability, as well as the measurement noise of the sensor. The matching module solves the inverse problem [8] by correlating the measured images to the nominal CAD data. The result of the matching process is the transformation of parameters between the two types of data with a measure of the goodness-of-fit. Once the matching parameters are known, decision criteria are employed to explore the options, such as whether to add a sensing point to the existing points or change the configurations of the sensor, i.e. to select a different combination of sensing points. The result of this planning process is the optimal configuration of the sensors.

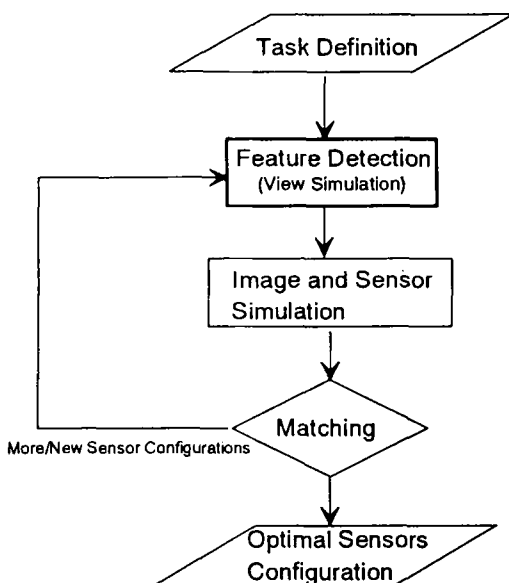


Fig. 1. Planning approach for laser scanners.

3. Object and Sensor Models

3.1 Laser Scanner Model

Exact acquisition of the geometrical characteristics and location of an object are among the most important factors in the success of an automatic assembly process. Currently, the geometrical acquisition sensors can be classified as performing geometrical dimensional inspections, or as being process-control oriented. The powerful sensors that fulfil these tasks are laser scanners, which operate on the triangulation principle [9, 10]. The advantage to profile scanning using laser sensors can be seen in the accurate acquisition of the coordinate data of the profile, making simultaneous derivation of the geometry, dimension, and location of the object possible. In addition, the fact that illumination is not required and that the system is internally calibrated is a great advantage and contributes to the easiest use of such sensors.

Laser scanners allow for non-tactile coordinate measurements of an object in one cross-section. In the subsequent task of sensor data processing, relevant data from a set of distance information are extracted. The relevant data vary depending on the task. For the assembly task, template matching to recognise the object and find its location is necessary. The result of profile data acquisition is comparable to the output of a vision system after application of edge-finding algorithms. The problem is to determine the position and geometry of an object from a set of measured coordinates.

A general description of a laser scanner is given in Fig. 2. The main optical characteristics are the field of view, depth of view, and the permissible inclination angle between the normal of the surface to the laser beam. All these characteristics form a set of constraints that determines the viewpoint location of the laser.

Three coordinate systems are considered in the scanning task (see Fig. 4): the laser coordinate system (Y_L, X_L, Z_L), the object coordinate system (X_o, Y_o, Z_o), and the world coordinate system (X_w, Y_w, Z_w). The laser viewpoint position is defined by the vector $r_L (X_L, Y_L, Z_L)$ in the object coordinate system, where the orientation (yaw and pitch) is defined by a unit vector along the bisector of the laser field of view cone, $p(\alpha, \beta)$. These five variables ($X_L, Y_L, Z_L, \alpha, \beta$)

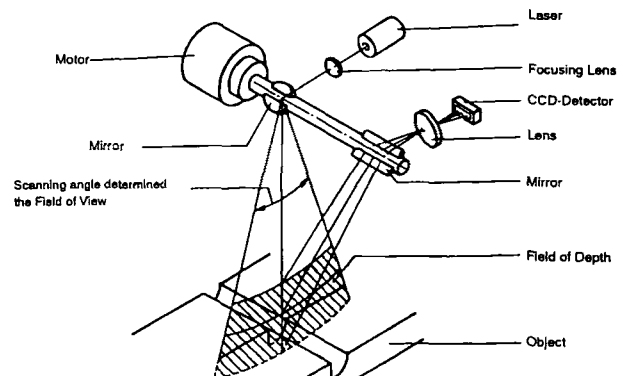


Fig. 2. General description of a laser scanner.

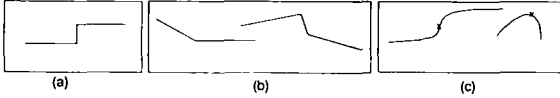


Fig. 3. A set of detectable features.

β) are planned in this work. The rotation around the laser beam (roll) is not considered since the 2D scanned feature (a profile) constrained it. The 3D location of the laser viewpoint is transformed to the world coordinate system in order to interact with the assembly robot sharing the same coordinate system.

3.2 Object Model

An object model contains geometric and topological data of the object as well as photometric properties. Attributes are attached to particular geometrical data defining critical assembly features. An assembly feature is defined as a region of interest within the object demonstrating critical assembly characteristics, such as edges of mating surfaces in mating objects [11]. The information about the critical features is used interactively in this work while planning the assembly task. A feature can be identified [12] by specific geometric characteristics such as the features shown in Fig. 3, i.e. (a) surface height function is discontinuous, (b) surface is continuous but the surface normal is discontinuous, (c) principal curvature is discontinuous and changes sign.

A 2D detectable feature and the different scanning parameters are shown in Fig. 4. Each feature consists of a set of discrete points $\{P_i, \dots, P_n\}$. The extreme points in the feature assisting in the acquisition process are $\{P_{mi}, P_{ma}, P_{mr}, P_{ml}\}$. Points P_{ml} , P_{mr} represent the start and the endpoint of the feature. Points P_{ma} , P_{mi} represent the extreme points in the feature relative to the laser viewpoint. These extreme points are determined as follows:

1. Connect by line points P_{mr} , P_{mi} .
The line forms an upper and lower convex half space, where the upper one is on the scanning side)
2. Select P_{ma} , a point from the upper half space with the maximal distance.
3. Select P_{mi} , a point from the lower half space with the maximal distance.

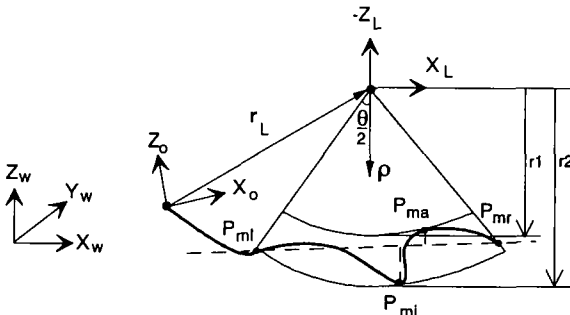


Fig. 4. A 2D detectable feature and the scanning parameters.

If more than one point of the feature meets the above conditions, select the point closest to the middle of the connecting line.

$P_i, \dots, P_n$	points along the scanning curve
P_{mi}	lowest point at the feature (point at the farthest location from the laser source)
P_{ma}	highest point at the feature (point at the nearest location from the laser source)
P_{ml}	start point of the feature
P_{mr}	last point of the feature
r_1, r_2	dimensions of the depth of view
(X_0, Y_0, Z_0)	object coordinate system
(X_L, Y_L, Z_L)	laser coordinate system
(X_w, Y_w, Z_w)	world coordinate system
$p(\alpha, \beta)$	a unit vector along the bisector of the laser field of view cone $p = [\sin(\beta)\cos(\alpha) \quad -\sin(\alpha) \quad \cos(\alpha)\cos(\beta)]$
θ	the scanning angle, θ_{max} defines the field of view
$r_L(X_L, Y_L, Z_L)$	the position vector of the origin of the laser beam

The photometric properties of an object are critical to the acquisition success. The variability of the surface appearance includes the colour, brightness and the roughness. The model in this work follows the detectability concept developed by Raczkowsky and Mittenbuehler [5], distinguishing between four important factors for successful application of optical sensors. These are: object geometry, surface properties, light source and technological properties of the sensor. Since laser scanners use active illumination techniques the light source influence can be neglected. Laser scanners use the diffuse reflection of the emitted laser beam to compute the distance of the object from geometrical relations (triangulation principle). Because the amount of scattered light is not constant a threshold is applied to the CCD data in the sensor. In current laser scanners the intensity information of the reflected light is not used to differentiate between the brightness levels and colours of the object. Thus, it is not necessary to fully model the intensity of the reflected light. For planning purposes, it is sufficient to distinguish between surfaces with a very high (mirror-like), very low (black) and average reflectivity. The roughness of the surface is modelled as a Gaussian noise and included in the image and sensor simulation stage (see Fig. 1).

4. Features Detectability Constraints

There is direct geometric relationship between the task requirements for scanning a surface and the set of laser locations satisfying these requirements. In general, each task requirement constrains a 3D region of acceptable location. The following three constraints affect the sensor location and

form a five-dimensional scanning space, where a point in this space is defined by $(X_L, Y_L, Z_L, \alpha, \beta)$.

Field of depth – all the profile points must lie within the range of acceptably sharp focus.

View angle – the maximum acceptable angle between the sensor beam and the surface normal.

Field of view – all the profile points must lie completely within the sensor field of view determined by the scanning angle.

4.1 Field of Depth (FOD) Analysis

The need for the scanned feature to be in focus constrains the location of the scanner. A point or part of the feature that is not in focus cannot be detected adequately. Even given that the focus distance is unique, there exists in current scanners a field of depth (FOD) characterising the distance between the farthest and the nearest object points for which the size of the blur is considered acceptable (e.g. not exceeding the maximal permissible light intensity). The FOD is the strip at distances r_1 and r_2 ($r_2 - r_1$) from the laser viewpoint. The following constraints have to be checked in order to verify that the feature is in the FOD:

$$(\mathbf{r}_{ma} - \mathbf{r}_L) \cdot \mathbf{p}' \geq r_1 \quad (1)$$

$$(\mathbf{r}_{mi} - \mathbf{r}_L) \cdot \mathbf{p}' \leq r_2 \quad (2)$$

where:

- $\mathbf{r}_L (X_L, Y_L, Z_L)$ the position vector of the laser viewpoint
- $\mathbf{r}_{mi} (X_{mi}, Y_{mi}, Z_{mi})$ the position vector of the farthest point of the feature
- $\mathbf{r}_{ma} (X_{ma}, Y_{ma}, Z_{ma})$ the position vector of the nearest point of the feature
- r_2, r_1 distances of FOD strip

The feasible region of the laser viewpoint position in the case of a 2D feature is shown in Fig. 5. It is generated by the intersection of two circles with radius r_1 and r_2 , where actually only the upper half space of the feature has to be considered.

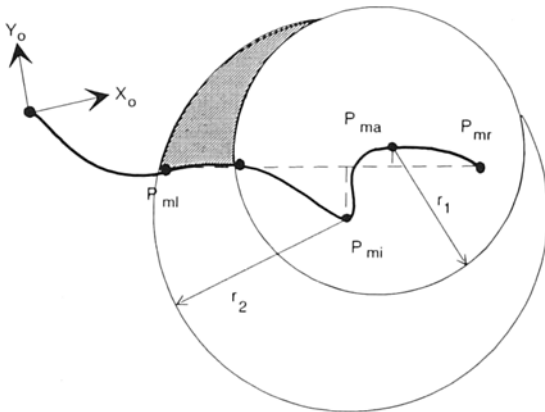


Fig. 5. The feasible position of the laser viewpoint under the field of depth constraint.

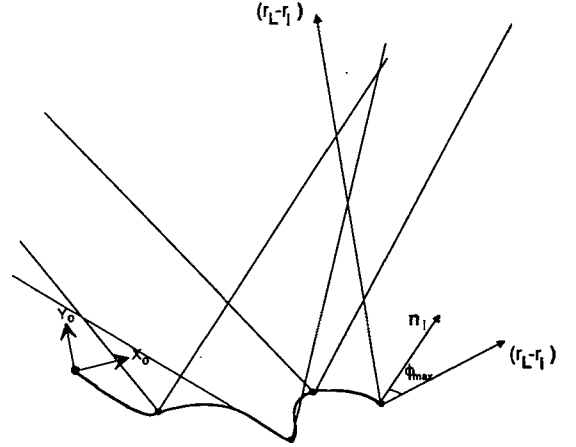


Fig. 6. The feasible position of the laser viewpoint for a 2D feature under the maximum view angle constraint.

4.2 View Angle Analysis

The view angle constraint is concerned with the maximum acceptable angle between the sensor beam and the surface normal. Owing to the optical system limitations the angle between the scanning beam and the normal to the surface (see Fig. 6) must be bounded, i.e. $\phi \leq \phi_{max}$. This constraint can be represented by a 3D cone whose apex is the respective point, axis is the normal to the surface, and viewing angle is $2\phi_{max}$. The constraint is determined as follows:

$$\frac{(\mathbf{r}_L - \mathbf{r}_i) \cdot \mathbf{n}_i'}{\|\mathbf{r}_L - \mathbf{r}_i\|} \leq \cos(\phi_{max}) \quad (3)$$

where:

- $\mathbf{r}_L (X_L, Y_L, Z_L)$ the position vector of the laser viewpoint
- $\mathbf{r}_i (X_i, Y_i, Z_i)$ the position vector of the point P_i of the feature
- $\mathbf{n}_i$ a unit vector of the surface normal at point P_i
- ϕ_{max} the maximum acceptable angle between the sensor beam and the surface normal

For a 2D feature this constraint forms a 2D viewing cone that characterizes the region of acceptable viewpoints. The cone apex is at the scanned point, its axis is the surface normal, $\mathbf{n}_i$, and its angle is $2\phi_{max}$. All the view angle constraints of the points along the scanned profile are merged by intersecting the respective viewing cones.

4.3 Field of View (FOV) Analysis

The field of view (FOV) constraint deals with the requirement that the feature to be scanned must lie completely within the FOV of the laser scanner. It forms a lower bound (a circular arc) on the distance between the scanned feature and the

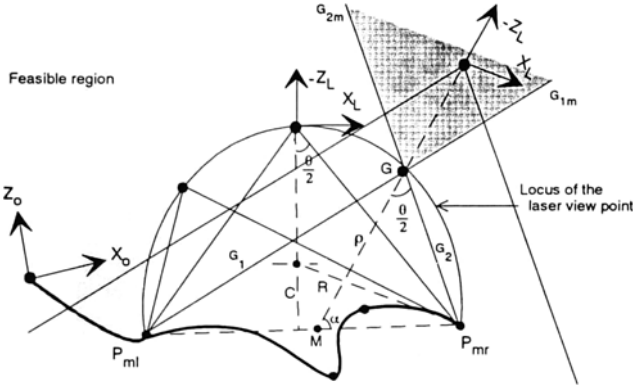


Fig. 7. The field of view constraint.

laser viewpoint (see Fig. 7). In order for the feature to be fully scanned the laser viewpoint must lie on the circular arc or outside of it. When the laser viewpoint lies on the bounding circular arc, the FOV of the laser scanner is bounded by the maximal scanning angle θ , and it can be represented as a 2D viewing cone. The apex of the cone is the laser viewpoint, its angle is θ , and it is spanned by the vectors to the start and the endpoints of the feature P_{ml} , P_{mr} . (For example, for the cone whose apex is the point G , spanned by G_1 and G_2 , $G = \{G_1, G_2\}$). The circular arc can be determined by its centre distance, C , from the line connecting P_{mr} and P_{ml} , and the radius, R , as follows [6]:

$$R = \frac{\bar{P}_{mr} - \bar{P}_{ml}}{2\sin(\theta)}, \quad C = \frac{\bar{P}_{mr} - \bar{P}_{ml}}{2\tan(\theta)} \quad (4)$$

The circular arc describes the locus of the laser viewpoint. Since the laser scanner can be tilted (around β , pitch) the actual bounding constraint is described by a sphere formed by the locus of the circular arc (see Fig. 8). For each viewpoint on this sphere there is one viewing direction, $\rho(\alpha, \beta)$. The viewing direction is the bisector of the viewing cone angle, θ . When the laser viewpoint lies outside the bounding sphere,

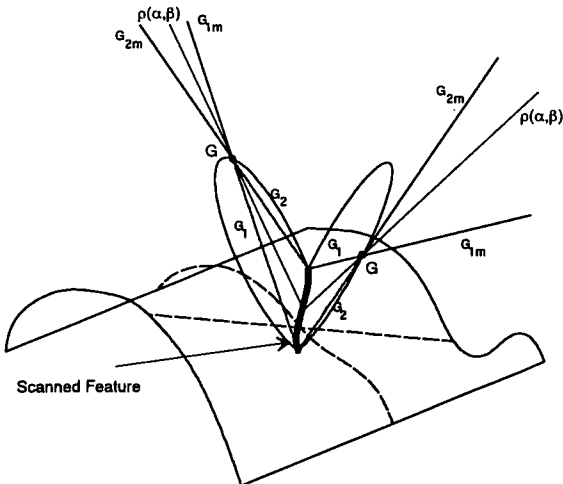


Fig. 8. Two examples of possible scanning location considering the field of view constraint.

a range of viewing directions can be obtained. In this case, the range of feasible laser viewpoint positions for a particular viewing direction, α , β , is represented by a 2D cone. The apex of this cone lies on the circular arc, its axis is the viewing direction (the opposite direction), its angle is equal to θ , and it is spanned by two half-lines forming the 2D viewing cone. (For example, Fig. 7 shows a 2D cone, $G = \{G_{1m}, G_{2m}\}$, its apex is G and it is spanned by G_{1m} and G_{2m}). Any point inside this cone satisfies a feasible laser viewpoint with particular orientation.

The constraint satisfying the locus of feasible laser viewpoint is actually the expression describing a general 2D cone with its apex on the bounding sphere (the locus of the circular arc), its axis opposite to the viewing direction, and its angle is θ . This expression is given by:

$$\frac{(\mathbf{r}_{Lp} - \mathbf{r}_G) \cdot (-\rho')}{\|\mathbf{r}_{Lp} - \mathbf{r}_G\|} \geq \cos\left(\frac{\theta_{max}}{2}\right) \quad (5)$$

where:

$\mathbf{r}_{Lp}(-X_L \cot(\beta), Y_L, X_L)$ position vector of the laser viewpoint on the plane at angle β formed by the circular arc and the scanned feature

θ scanning angle, θ_{max} defines the field of view

$\rho(\alpha, \beta)$ $\rho = [\sin(\beta)\cos(\alpha) - \sin(\alpha)\cos(\alpha)\cos(\beta)]$

$\mathbf{r}_M(X_M, Y_M, Z_M)$ position vector of the intersection of the viewing direction vector and the line connecting P_{ml} and P_{mr}

R radius of the circumscribed circle $R = (\bar{P}_{mr} - \bar{P}_{ml})/2\sin(\theta)$

$\overline{MG}$ distance between the point M and G

$$\overline{MG} = \frac{2R\left(\sin\left(\alpha - \frac{\theta}{2}\right)\sin\left(\alpha + \frac{\theta}{2}\right)\right)}{\sin(\alpha)}$$

$\mathbf{r}_G(X_G, Y_G, Z_G)$ position vector of a laser viewpoint lying on the circular curve

$$\mathbf{r}_G = \mathbf{r}_M + \mathbf{r}_{MG}; \quad \mathbf{r}_{MG} = \overline{MG} \cdot \rho$$

5. Constraints Integration

Following the idea of a synthesis approach each task constraint has been analytically modelled independently of the other constraints. At this point, all these constraints are to be combined in order to find a set of parameters that satisfy all of them simultaneously. Merging the constraints is formulated as a constrained optimisation problem. All three constraints: field of view, depth of view, and view of angle must share an admissible domain. Consequently, in order to determine admissibility of viewpoint with respect to them it is desired to evaluate the following relationships:

$$g_{1a} = (\mathbf{r}_{mu} - \mathbf{r}_L) \cdot \mathbf{p}' \geq r_1 \quad (6)$$

$$g_{1b} = (\mathbf{r}_{mi} - \mathbf{r}_L) \cdot \mathbf{p}' \leq r_2 \quad (7)$$

$$g_2 = \frac{(\mathbf{r}_L - \mathbf{r}_i) \cdot \mathbf{n}_i'}{\|\mathbf{r}_L - \mathbf{r}_i\|} \leq \cos(\phi_{max}) \quad \text{for } i = 1, \dots, n \quad (8)$$

$$g_3 = \frac{(\mathbf{r}_{Lp} - \mathbf{r}_G) \cdot (-\mathbf{p}')}{\|\mathbf{r}_{Lp} - \mathbf{r}_G\|} \geq \cos\left(\frac{\theta_{max}}{2}\right) \quad (9)$$

The view of angle constraint (g_2) must be resolved for all the n points along the feature, while for the depth of view and field of view constraints, there is a single constraint for each feature (g_{1a} , g_{1b} , g_3).

Although the above constraints address an admissibility solution of the laser scanner viewpoint, a cost function is defined in order to find the best viewpoint. The cost function that was selected is the sum of the view angles. Minimisation of this function gives a solution which guarantees more robust scanning, since ideal scanning position is orthogonal to each point in the surface, i.e. when $\phi=0$. The cost function is taken to be a minimisation of all the view angles of the points along the scanned feature. Thus, the optimisation problem is written as follows:

Find (X_L , Y_L , Z_L , α , β) such that:

$$\text{Min} \left(\sum_{i=1}^n \phi_i \right)$$

subject to:

$$g_{1a}, g_{1b}, g_2, g_3$$

As an initial viewpoint to start the optimisation process the view angle constraint is applied. By using a CAD model the respective view cones along the features are intersected, resulting in a feasible viewing domain. The optimisation initial condition is selected from this domain. This initial point guarantees fast convergence of the optimisation process. Given the above formulation the optimisation starts with an initial point in the domain of possible viewpoints, and then generates a viewpoint that is globally admissible and locally optimal following the optimisation procedure.

6. Examples

In this section, two examples are presented to demonstrate the sensor planning approach. The previous planning algorithms for features detectability satisfaction are used and show their effectiveness by planning a robotic set-up. In the first example, the planning algorithms are applied to scanning a 2D feature resulting in the optimal viewpoint of the laser scanner. In the second example, the planning algorithms are applied to assist in the task of the car door assembly. The above techniques are applied to the robotic set-up in order to find optimal viewpoints in the critical assembly features.

6.1 An Example of Viewpoint Planning for a 2D Feature

In the following example, a 2D feature is considered (see Fig. 9). The goal is to find an optimal laser viewpoint (X_L ,

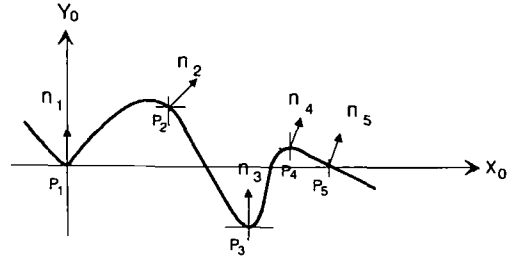


Fig. 9. A 2D feature.

Y_L , α) satisfying the feature detectability constraints. For reasons of simplicity only five points (selected arbitrarily) along the feature are taken into account $\{P_1, P_2, P_3, P_4, P_5\}$.

The normal and position vectors of the points are:

$$\begin{aligned} \mathbf{n}_1 &= 0i + 1j & \mathbf{r}_L &= x_L i + y_L j \\ \mathbf{n}_2 &= 0.7071i + 0.7071j & \mathbf{r}_{mi} = \mathbf{r}_{P3} &= 4.5i - 1.5j \\ \mathbf{n}_3 &= 0i + 1j & \mathbf{r}_{ma} = \mathbf{r}_{P3} &= 2.5i + 1.5j \\ \mathbf{n}_4 &= 0.316i + 0.95j & \mathbf{r}_{mi} = \mathbf{r}_{P1} &= 0 \\ \mathbf{n}_5 &= 0.316i + 0.95j & \mathbf{r}_{mr} = \mathbf{r}_{P5} &= 6.5i + 0j \end{aligned}$$

The laser scanner parameters are:

$$\begin{aligned} r_1 &= 30 \text{ mm}, \quad r_2 = 65 \text{ mm} \\ \text{FOD } r_2 - r_1 &= 35 \text{ mm} \\ \theta_{max} &= 80^\circ \text{ (max. FOV angle)} \\ \phi &= \pm 45^\circ \text{ (max. permissible view angle)} \end{aligned}$$

With the above data, the objective function and the optimisation constraints (given by equations (6)–(9) are constructed. The optimisation is performed using the MatLab constrained optimisation routine CONSTR [13]. This routine uses a sequential quadratic programming method to solve the general nonlinear problem.

The example has been realised on a 386 micro-processor system (25 MHz) equipped with a mathematical coprocessor. The planning time took 3.5 s (20116 flops).

The domain satisfying a feasible position of the laser viewpoint is shown in Fig. 10. It is bounded by the extreme points view angles (i.e. P_{mi} , P_{mr}), and the depth of view constraint, maximal distance (r_2). The optimal laser viewpoint obtained after the optimisation process is:

$$X_L = 43.2 \text{ mm}, \quad Y_L = 43.2 \text{ mm}, \quad \alpha = 5.0^\circ$$

6.2 Laser Viewpoint Planning for Assisting Assembly Task

In the following example, laser viewpoint locations are planned and integrated into an assembly task. The task is to assemble a car door into the body in white (BIW). During the final stage of the BIW manufacturing, major panels, such as doors are attached to the BIW openings with hanging fixtures. Using a door as an example, the door is first fitted to the body side opening with a hanging fixture. In case gap and flush do not meet the dimensional requirement upon visual inspection, the door is further adjusted manually by moving the hinge or bending and twisting the door. The problem

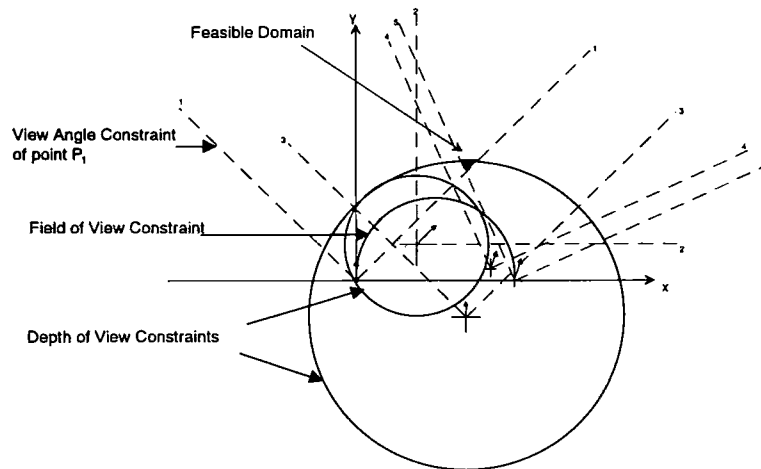


Fig. 10. The feature detectability viewpoint position constraints.

with current panel assembly in the automotive industry is that a systematic procedure is not available, and the way in which sensing elements, robots, and further automation should be integrated is not clear. In some exceptional cases in which sensors were installed, determination of a suitable viewpoint has to be found by a time-consuming generate and test approach.

In this example, the previous laser-scanning planning algorithms are applied to the car door assembly task, where the aim is to automate robotic assembly. The car door is assembled by a robot and assisted by a laser scanner sensor that is fastened to it. The optimal laser viewpoints are planned in advance and helped in forming the robotic set-up as a part of the off-line programming of the robot. This example demonstrates the effectiveness of the planning approach as well as the advantages of using a laser scanner sensor. Planning the laser viewpoints in advance saves the tedious iterations of manipulating the sensor while it is fastened to the robot. The laser scanner sensor is proved to be attractive for assisting an assembly task mainly because of the simplicity of the data acquisition and processing.

Experimental Set-up

The experimental set-up is shown in Fig. 11. A six degrees of freedom robot (ABB 3000) is assisting in manipulating the door and fitting it to the BIW. A vacuum gripper is attached to the robot for the handling task. A laser scanner is fastened to the robot gripper. The robot can manipulate the laser scanner and locate the sensor in the planned viewpoint locations. These locations are the result of the previous planning algorithms and the detectability constraints.

The laser scanner parameters are:

$$r_1 = 135 \text{ mm}, \quad r_2 = 165 \text{ mm}$$

$$\text{FOD } r_2 - r_1 = 30 \text{ mm}$$

$$\theta_{\max} = 32.5^\circ \text{ (max. FOV angle)}$$

$$\phi = \pm 45^\circ \text{ (max. permissible view angle)}$$

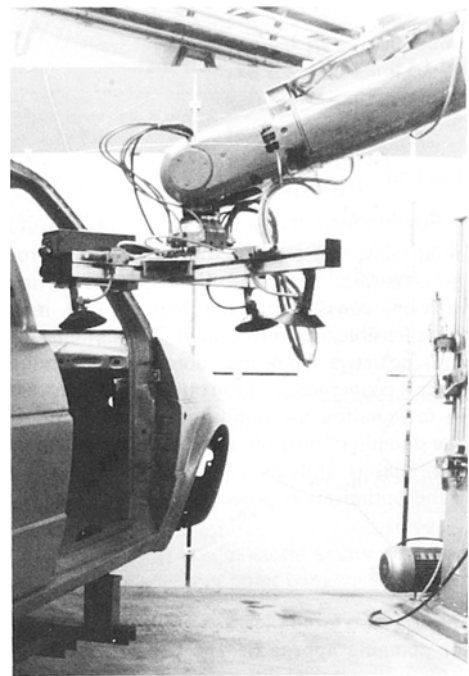


Fig. 11. The robotic cell for automatic assembly of car doors.

Laser-scanner Planning Results

At first the critical assembly features are identified to make the assembly process reliable. The geometrical data of the task is represented in the CATIA – Solid Modeller environment (IBM 4381). The solid modeller provides a programming environment which facilitates the designer interaction as well as determining the initial point for the optimisation of the optimal laser scanner viewpoint. Determining feasible regions that satisfy the view angle constraints takes about 3 s on the IBM 4381. The program was written in FORTRAN 77 using the CATIA geometric functions. In this experiment five

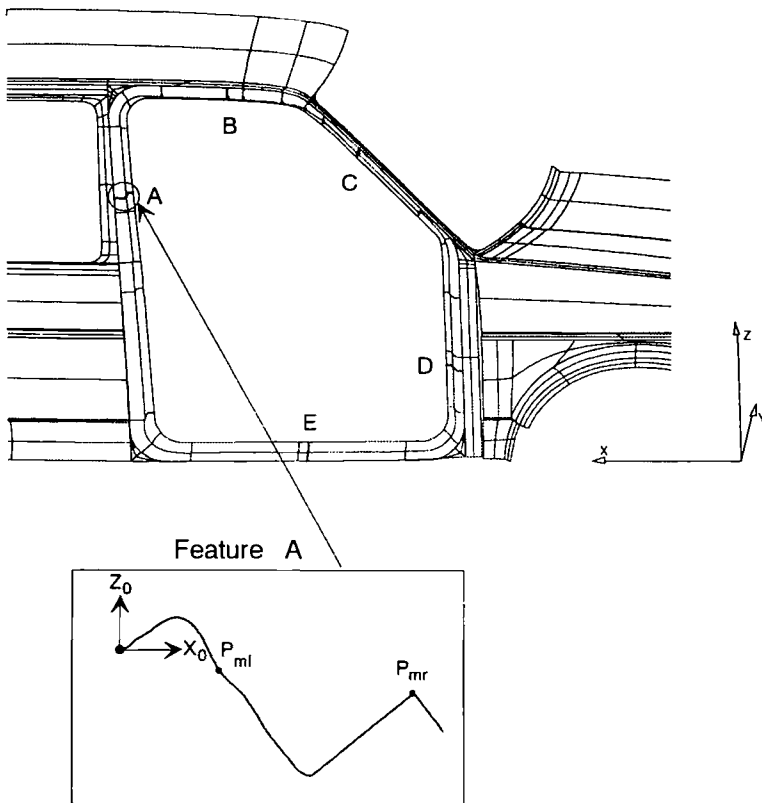


Fig. 12. The car door-opening and its respective assembly features.

features were selected in the car door-opening $\{A, B, C, D, E\}$, (see Fig. 12). Once the assembly features have been marked on the CAD car model the feature detection planning process can be undertaken. The aim is to find the optimal viewpoint of the laser scanner for each one of the selected features at the BIW.

As an example for planning the optimal viewpoint, feature A at the left-hand side pillar of the car door-opening is selected. The optimisation was performed by the same computing environment as in the first example. The initial starting point for the optimisation was selected from the feasible regions computed in CATIA. The planning time took 12 s (41213 flops) and resulted in the following location:

$$X_L = 45.5 \text{ mm}, \quad Y_L = 0.0 \text{ mm}, \quad Z_L = 92.4 \text{ mm}, \\ \alpha = 5.0^\circ, \quad \beta = 0.0^\circ$$

7. Summary and Conclusions

An automatic planning system to determine the optimal location of sensor viewpoints, in this case using laser scanner sensors, was presented. The planning system determines the viewpoint by an optimisation approach, where the constraints are: the depth of view, field of view, and maximal view angle of the sensor. The resultant sensor viewpoint satisfied the above scanned feature detectability constraints and guaranteed global scanning of the feature.

The experiment using the task of car door assembly demonstrated the significance of the sensor planning approach as well as the advantages of the laser scanner for assisting in automatic assembly tasks.

This planning approach can assist with the fast implementation of automatic assembly cells integrated with sensors, since the set-up time of the sensors will be shortened significantly. In the future this sensor planning approach is to be integrated into a robot off-line programming system in order to form a complete system that supports sensor-aided automatic assembly tasks.

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